

Ex No :	Class Component
Date :	

AIM:

To write and implement a program on React using Class component.

A. Scenario: Build a remote learning application where teachers can track students attendance in a live session.

Tasks:

1. Create a StudentCard class component that:
 - Displays a student's name, attendanceStatus (e.g., "present", "absent")
 - joinTime as props.
 - Changes the background color to green for "present" and red for "absent".
2. Add a "Mark Present" button that updates the attendanceStatus to "present" and logs "Student [Name] is now marked as present."
3. If the joinTime is after 10:00 AM, display a warning saying "Late Joiner".

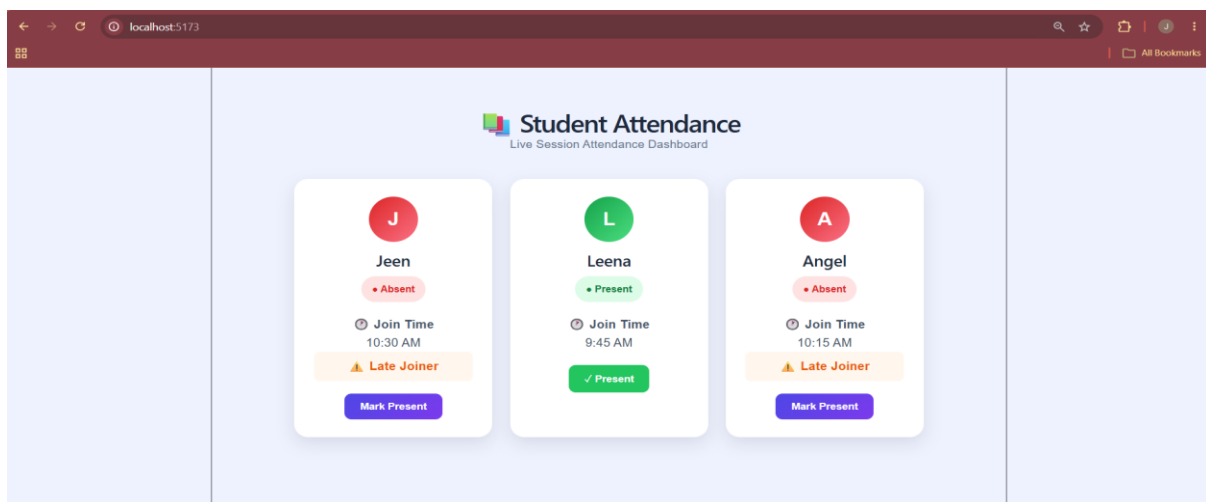
PROGRAM:**App.jsx:**

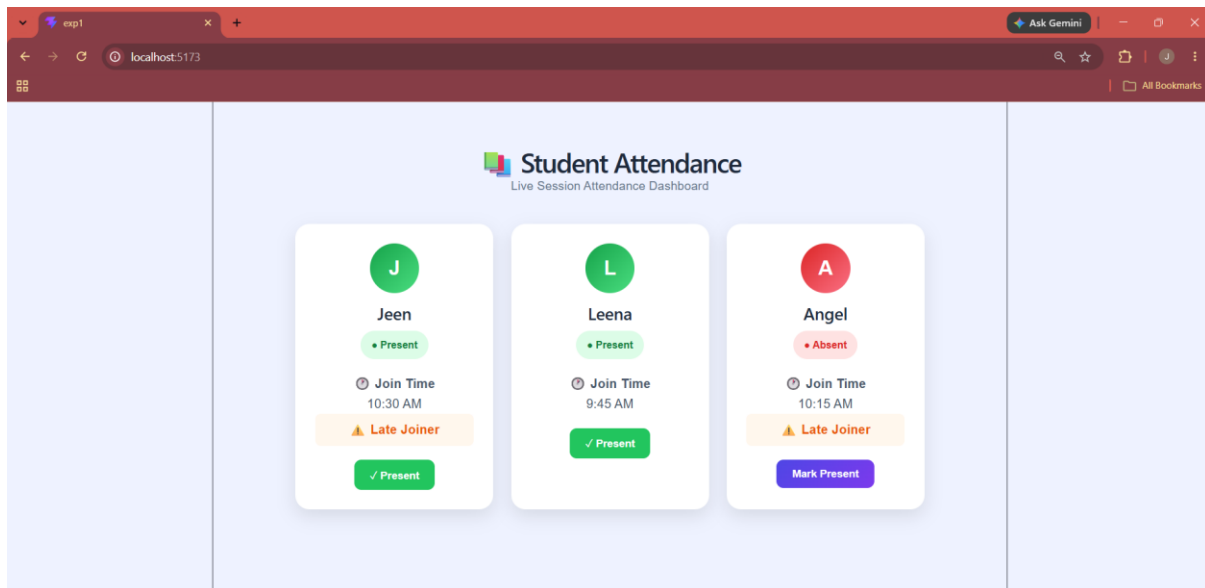
```
import React from "react";
import "./App.css";
class StudentCard extends React.Component {
  constructor(props) {
    super(props);
    this.state = {
      attendanceStatus: props.attendanceStatus
    };
  }
  markPresent = () => {
    this.setState({
      attendanceStatus: "present"
    });
  }
}
```

```
});  
console.log(  
  `Student ${this.props.name} is now marked as present.`  
);  
};  
  
isLate = () => {  
  const [time, period] = this.props.joinTime.split(" ");  
  let [hour, minute] = time.split(":").map(Number);  
  if (period === "PM" && hour !== 12) {  
    hour += 12;  
  } if (period === "AM" && hour === 12) {  
    hour = 0;  
  } return hour > 10 || (hour === 10 && minute > 0);  
};  
  
render() {  
  const { name, joinTime } = this.props;  
  const { attendanceStatus } = this.state;  
  const isPresent = attendanceStatus === "present";  
  return (  
    <div className="student-card">  
      <div className={`student-icon ${isPresent ? "green" : "red"} `}>  
        {name.charAt(0)}  
      </div>  
      <h2>{name}</h2>  
      <span className={`status ${isPresent ? "present" : "absent"} `}>  
        {isPresent ? "● Present" : "● Absent"}  
      </span>  
      <div className="student-info">  
        <p>
```

```
<b>🕒 Join Time</b>
<br />
{joinTime}
</p>
{this.isLate() && (
  <p className="late-warning">
    ⚠ Late Joiner
  </p>
)}
</div>
<button
  className="present-btn"
  onClick={this.markPresent}
  disabled={isPresent}
>{isPresent ? "✓ Present" : "Mark Present"}
</button>
</div>
);
}
}
function App() {
  return (
    <div className="attendance-app">
      <div className="header">
        <h1>📚 Student Attendance</h1>
        <p>Live Session Attendance Dashboard</p>
      </div>
      <div className="students-container">
        <StudentCard
```

```
        name="Jeen"  
        attendanceStatus="absent"  
        joinTime="10:30 AM"  
    />  
  
    <StudentCard  
        name="Leena"  
        attendanceStatus="present"  
        joinTime="9:45 AM"  
    />  
  
    <StudentCard  
        name="Angel"  
        attendanceStatus="absent"  
        joinTime="10:15 AM"  
    />  
  
    </div>  
  
</div>  
  
);  
}  
  
export default App;
```

OUTPUT:



B. Emergency Alert System using class component

Scenario: Build a city-wide emergency alert system where alerts are dynamically displayed based on severity.

Tasks:

1. Create an AlertCard class component that:
 - Accepts title, message, and severity as props. Severity can be "low", "moderate", or "high".
 - Dynamically assigns a background color (e.g., green for "low", orange for "moderate", red for "high").
2. Add a "Snooze Alert" button that hides the alert for 10 seconds and then displays it again (use setTimeout and state).
3. If the message exceeds 150 characters, display a "Show More" button that toggles the full text when clicked.

PROGRAM:

App.jsx:

```
import React from "react";
import "./App.css";
class AlertCard extends React.Component {
  constructor(props) {
    super(props);
    this.state = {
```

```
    visible: true,
    showFull: false
  };
}

snoozeAlert = () => {
  this.setState({
    visible: false
  });
  setTimeout(() => {
    this.setState({
      visible: true
    });
  }, 10000);
};

toggleText = () => {
  this.setState({
    showFull: !this.state.showFull
  });
};

render() {
  const { title, message, severity } = this.props;
  const { visible, showFull } = this.state;
  if (!visible) {
    return (
      <div className="snoozed-card">
        <span>🔔</span>
        <h3>Alert Snoozed</h3>
        <p>Alert will appear again in 10 seconds.</p>
      </div>
    );
  }
}
```

```
);  
}  
let icon;  
let className;  
if (severity === "low") {  
    icon = "✓";  
    className = "low";  
} else if (severity === "moderate") {  
    icon = "⚠";  
    className = "moderate";  
} else {  
    icon = "🔥";  
    className = "high";  
}  
const shortMessage = message.substring(0, 150);  
return (  
    <div className={`alert-card ${className}`}>  
        <div className="alert-top">  
            <div className="alert-icon">  
                {icon}  
            </div>  
            <div>  
                <h2>{title}</h2>  
                <span className="severity">  
                    {severity.toUpperCase()}  
                </span>  
            </div>  
        </div>  
        <p className="alert-message">
```

```

    {showFull || message.length <= 150
      ? message
      : shortMessage + "..."}
  </p>

  {message.length > 150 && (
    <button className="show-btn" onClick={this.toggleText} >
      {showFull ? "Show Less ▲" : "Show More ▼"}
    </button>
  )}

  <button className="snooze-btn" onClick={this.snoozeAlert}
> 🛑 Snooze Alert
</button>
</div>

);
}}

function App() {
  return (
    <div className="emergency-app">
      <div className="emergency-header">
        <div className="city-icon"> 🚨 </div>
        <h1>Emergency Alert Center</h1>
        <p>City-wide emergency notifications</p>
      </div>
      <div className="alerts-container">
        <AlertCard
          title="Flood Warning"
          severity="high"

```

```
message="Heavy rainfall is expected across the city. Citizens living in low-lying areas are advised to move to safer locations immediately. Avoid flooded roads and follow instructions from emergency authorities."/> <AlertCard
```

```
title="Traffic Advisory"
```

```
severity="moderate"
```

```
message="Heavy traffic has been reported on major roads. Please use alternative routes."
```

```
/>
```

```
<AlertCard
```

```
title="Weather Update"
```

```
severity="low"
```

```
message="Weather conditions are currently normal. No immediate action is required."
```

```
/>
```

```
</div>
```

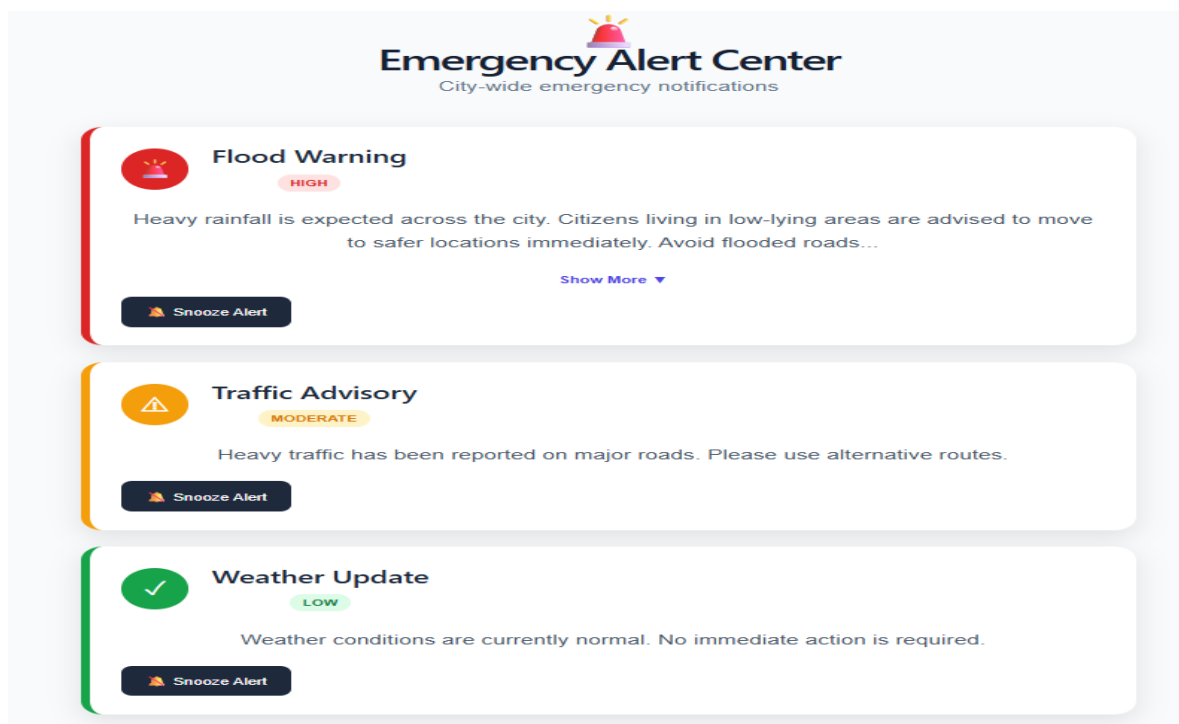
```
</div>
```

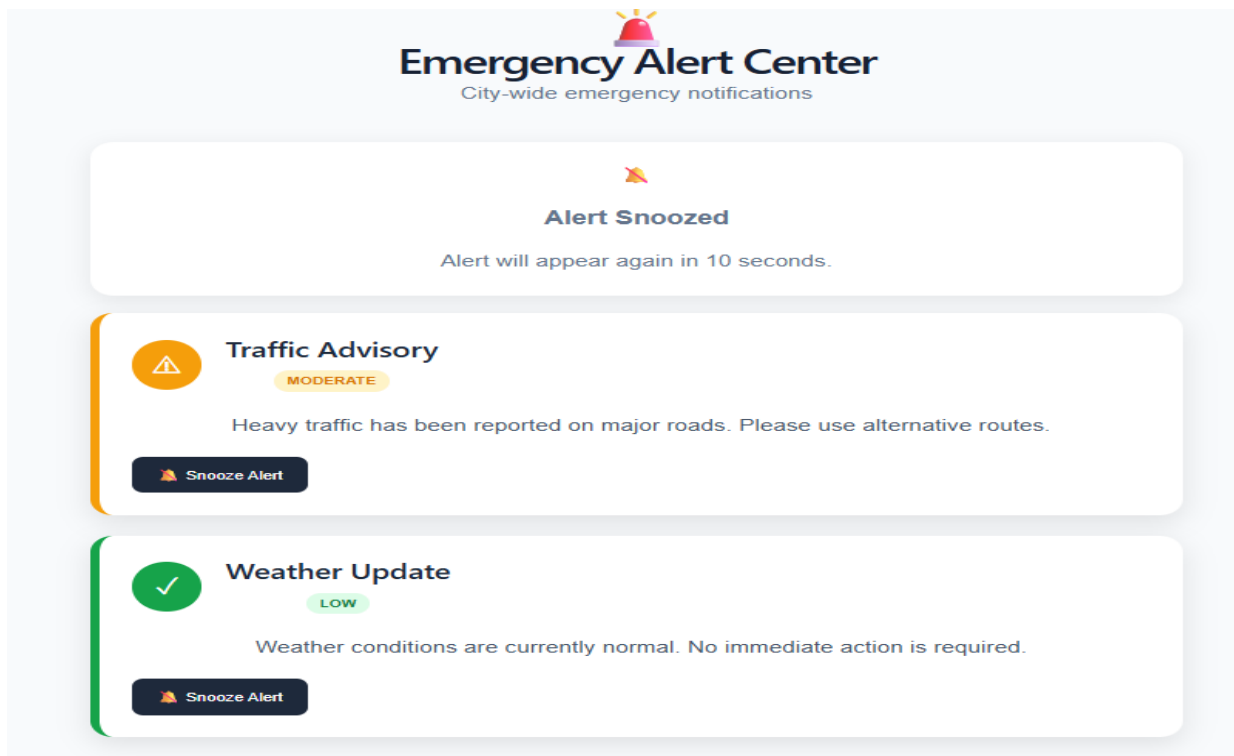
```
);
```

```
}
```

```
export default App;
```

OUTPUT:





3. Print Name on Button Click

PROGRAM:

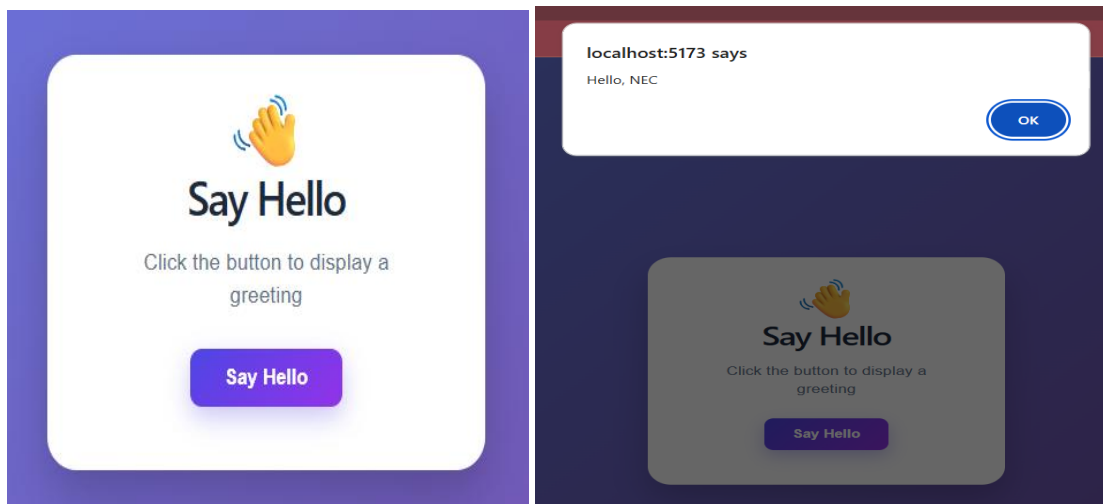
App.jsx:

```
import React from "react";
import "./App.css";
const App = () => {
  const handleClick = (name) => {
    alert(`Hello, ${name}`);
  };
  return (
    <div className="hello-page">
      <div className="hello-card">
        <div className="hello-icon">👋</div>
        <h1>Hello Application</h1>
        <p>
          Click the button to display a greeting.
        </p>
```

```

    <button onClick={() => handleClick("NEC")}>
      Say Hello
    </button>
  </div>
</div>
);
};
export default App;

```

OUTPUT:**D. Multiple Components****Use Header, Footer and Body components****PROGRAM:****App.jsx:**

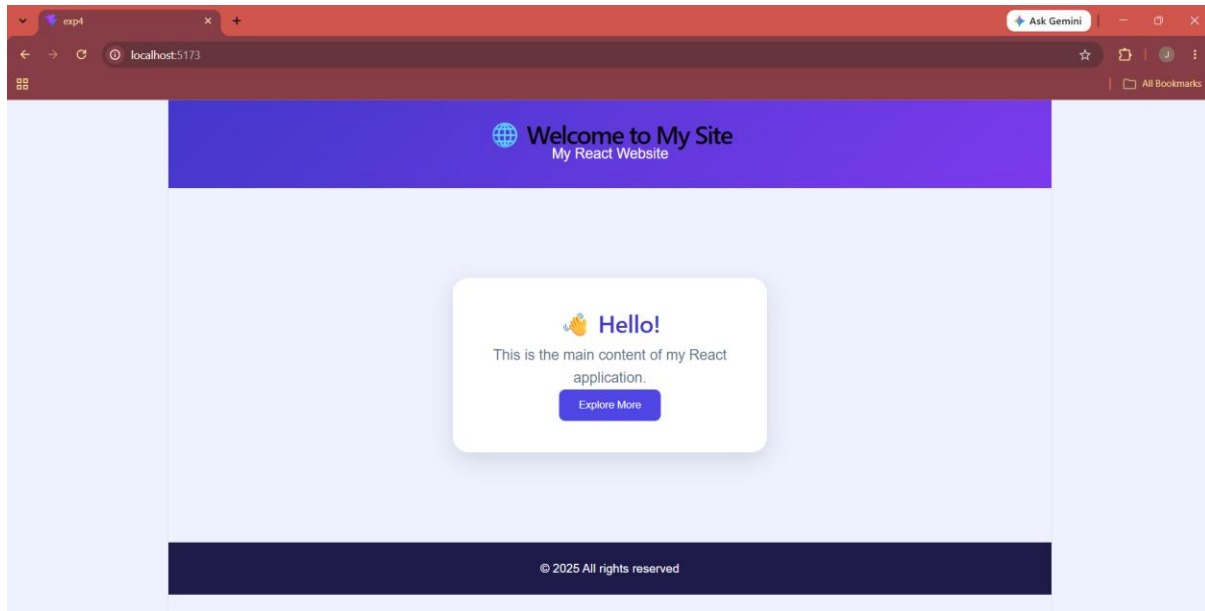
```

import React from "react";
import "./App.css";
const Header = () => {
  return (
    <header>
      <h1>🌐 Welcome to My Site</h1>
      <p>My React Website</p>
    </header>
  )
}

```

```
);  
};  
const Body = () => {  
  return (  
    <main>  
      <div className="content-card">  
        <h2>👋 Hello!</h2>  
        <p>  
          This is the main content of my React application.  
        </p>  
        <button>Explore More</button>  
      </div>  
    </main>  
  );  
};  
const Footer = () => {  
  return (  
    <footer>  
      <small>© 2025 All rights reserved</small>  
    </footer>  
  );  
};  
const App = () => {  
  return (  
    <>  
      <Header />  
      <Body />  
      <Footer />  
    </>  
  );  
};
```

```
);  
};  
  
export default App;
```

OUTPUT:

Problem Understanding & Approach (10)	Implementation & Execution (15)	Tool/Technology Usage (10)	Code Quality & Documentation (5)	Viva (5)	Time Management (5)	Total (50)

Result:

Thus, the Program on React using class component is completed and the output is verified.